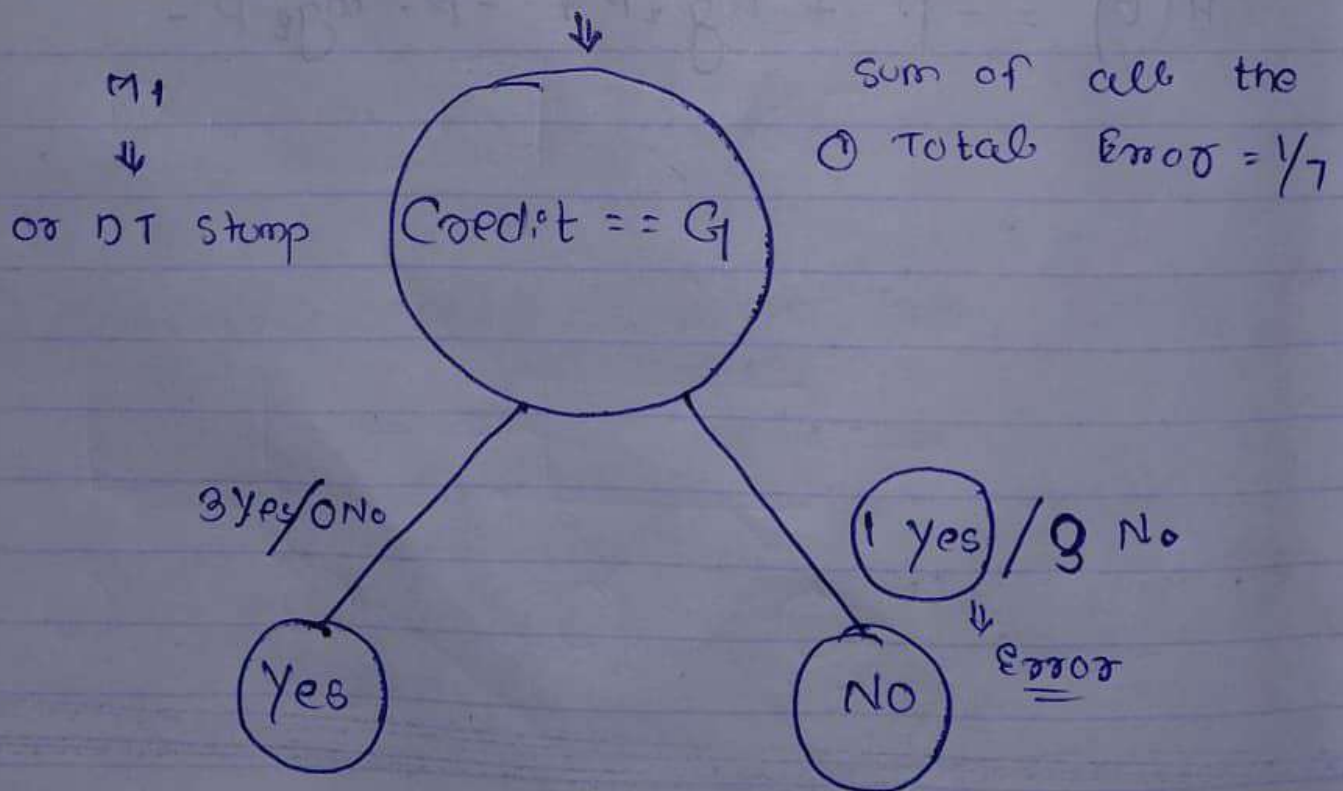


② Sum of the Total Error And Performance of Stump

<u>Salary</u>	<u>Credit</u>	<u>Approval</u>	<u>Sample weight</u>
$\leq 50K$	B	No	$1/7$
$\leq 50K$	G	Yes	$1/7$
$\leq 50K$	G	Yes	$1/7$
$> 50K$	B	No	$1/7$
$> 50K$	G	Yes	$1/7$
$> 50K$	N	Yes	$1/7$
$\leq 50K$	N	No	$1/7$



② Performance of stamp $= \frac{1}{2} \ln \left[\frac{1 - \tau_E}{\tau_E} \right]$

$$= \frac{1}{2} \ln \left[\frac{1 - 1/7}{1/7} \right]$$

$$= \frac{1}{2} \ln \left[\frac{7/7 - 1/7}{1/7} \right]$$

$$= \frac{1}{2} \ln \left[\frac{6/7}{1/7} \right]$$

$$= \frac{1}{2} \ln \left[\frac{6}{1} \times \frac{7}{1} \right]$$

$$= \frac{1}{2} \ln [6]$$

$$= \frac{1}{2} \log_e (6)$$

$$\text{or } \log_e (6)$$

$$\therefore \log_e (6) \approx 1.791759$$

$$= \frac{1}{2} \times 1.791759$$

$$= 0.8958795 \approx \underline{\underline{0.896}}$$

or

$$\frac{1}{2} \ln \left(\frac{1 - 1/7}{1/7} \right) = \frac{1}{2} \ln(6) = \frac{1}{2} (1.791759) \approx \underline{\underline{0.896}}$$

Performance of stump ≈ 0.896

↓

$$f = \alpha_1 (m_1) + \alpha_2 (m_2) + \dots + \alpha_n (m_n)$$

$$\boxed{\alpha_1 = 0.896} \rightarrow \text{weight}$$

